

✓ Quiz 4

This is an individual assignment.

Write your answers using markdown cells or embed handwritten answers with

`IPython.display.Image`.

✓ Exercise 1 (6 points)

Complete the following table for the classifiers listed below.

Under "Type" list: generative, discriminative or probabilistic discriminative.

Under "Convergence" list whether the convergence is guaranteed or any other conditions for the converged solution.

Classifier	Type	Mapper function	Objective Function	Learning Algorithm	Assumptions
Naïve Bayes Classifier					
Fisher's LDA					
Perceptron					
Logistic Regression					
Hard-margin SVM					
Soft-margin SVM					

Answer:

Classifier	Type	Mapper function	
Naïve Bayes Classifier	Generative	$P(y x) \propto P(x y)P(y)$	Maximum Likelihood Estimation
Fisher's LDA	Discriminative	$w^T x$ (linear projection)	Maximize $\frac{w^T S_B w}{w^T S_W w}$ (between-class variance to within-class variance)
Perceptron	Discriminative	$\text{sign}(w^T x + b)$	Minimize misclassification
Logistic Regression	Probabilistic Discriminative	$\sigma(w^T x + b)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$	Minimize negative log-likelihood
Hard-margin SVM	Discriminative	$\text{sign}(w^T x + b)$	Minimize $\frac{1}{2} w ^2$ subject to margin constraints
Soft-margin SVM	Discriminative	$\text{sign}(w^T x + b)$	Minimize $\frac{1}{2} w ^2 + \text{margin violation}$

✓ Exercise 2 (3 points)

Consider the data matrix $\mathbf{X}$ of size $3 \times N$, where N is the number of samples.

The covariance matrix $\mathbf{K}$, of size 3×3 has 3 eigenvectors

$v_1 = [-0.99, 0.09, 0]^T$, $v_2 = [0, 0, 1]^T$ and $v_3 = [-0.09, -0.99, 0]^T$ with eigenvalues $\lambda_1 = 0.98$, $\lambda_2 = 0.5$ and $\lambda_3 = 1.98$, respectively. Answer the following questions:

1. (1 point) **What linear transformation would you use to uncorrelate the data $\mathbf{X}$? Provide a numerical solution with the information provided. Justify your answer.**
2. (1 point) **Define the linear transformation (with all numerical entries) that Principal Component Analysis (PCA) would use to project this 3-dimensional data to a 2-dimensional representation.**
3. (0.5 points) **What is the amount of explained variance of this 2-D projection? Justify.**
4. (0.5 points) **Let $\mathbf{Y}$ be the data (linear) transformation obtained by principal component transform of $\mathbf{X}$ onto a 2-dimensional space. What is resulting covariance matrix of transformed data $\mathbf{Y}$? Use a numerical answer and justify your answer.**

✓ Answer:

1. Linear transformation to uncorrelate the data (1 point)

To uncorrelate the data $\mathbf{X}$, we need to diagonalize the covariance matrix $\mathbf{K}$. This is achieved by transforming the data using the eigenvector matrix.

Transformation: $\mathbf{Z} = \mathbf{V}^T \mathbf{X}$

where $\mathbf{V}$ is the matrix whose columns are the eigenvectors:

$$\mathbf{V} = \begin{bmatrix} -0.99 & 0 & -0.09 \\ 0.09 & 0 & -0.99 \\ 0 & 1 & 0 \end{bmatrix}$$

Therefore, the transformation matrix is:

$$\mathbf{V}^T = \begin{bmatrix} -0.99 & 0.09 & 0 \\ 0 & 0 & 1 \\ -0.09 & -0.99 & 0 \end{bmatrix}$$

Justification: The eigenvector matrix $\mathbf{V}$ diagonalizes the covariance matrix (since $\mathbf{K} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ where $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues). After transformation, the covariance matrix of $\mathbf{Z}$ becomes $\mathbf{\Lambda}$, which is diagonal, meaning the features are uncorrelated.

2. PCA transformation to 2D (1 point)

For PCA, we select the top $k = 2$ eigenvectors corresponding to the largest eigenvalues.

Eigenvalues in descending order:

- $\lambda_3 = 1.98$ (largest) $\rightarrow$ eigenvector $v_3 = [-0.09, -0.99, 0]^T$
- $\lambda_1 = 0.98$ (second largest) $\rightarrow$ eigenvector $v_1 = [-0.99, 0.09, 0]^T$
- $\lambda_2 = 0.5$ (smallest) $\rightarrow$ eigenvector $v_2 = [0, 0, 1]^T$

The PCA transformation matrix $\mathbf{W}$ is formed by taking the top 2 eigenvectors as columns:

$$\mathbf{W} = [v_3 \quad v_1] = \begin{bmatrix} -0.09 & -0.99 \\ -0.99 & 0.09 \\ 0 & 0 \end{bmatrix}$$

The linear transformation is: $\mathbf{Y} = \mathbf{W}^T \mathbf{X}$

where:

$$\mathbf{W}^T = \begin{bmatrix} -0.09 & -0.99 & 0 \\ -0.99 & 0.09 & 0 \end{bmatrix}$$

This projects the 3D data $\mathbf{X}$ onto a 2D space $\mathbf{Y}$.

✓ 3. Explained variance of 2D projection (0.5 points)

Total variance = Sum of all eigenvalues =

$$\lambda_1 + \lambda_2 + \lambda_3 = 0.98 + 0.5 + 1.98 = 3.46$$

Variance retained by top 2 components $= \lambda_3 + \lambda_1 = 1.98 + 0.98 = 2.96$

Explained variance ratio $= \frac{2.96}{3.46} = 0.8555$ or **85.55%**

Justification: The explained variance is calculated as the ratio of the sum of the eigenvalues corresponding to the selected principal components to the sum of all eigenvalues. Since we selected the top 2 eigenvectors (with eigenvalues 1.98 and 0.98), we retain 85.55% of the total variance in the original data.

4. Covariance matrix of transformed data Y (0.5 points)

When we apply the PCA transformation $\mathbf{Y} = \mathbf{W}^T \mathbf{X}$, the covariance matrix of $\mathbf{Y}$ becomes:

$$\text{Cov}(\mathbf{Y}) = \mathbf{W}^T \mathbf{K} \mathbf{W}$$

Since $\mathbf{W}$ contains the eigenvectors of $\mathbf{K}$, and the eigenvectors are orthogonal, the resulting covariance matrix is diagonal with the corresponding eigenvalues:

$$\text{Cov}(\mathbf{Y}) = \begin{bmatrix} \lambda_3 & 0 \\ 0 & \lambda_1 \end{bmatrix} = \begin{bmatrix} 1.98 & 0 \\ 0 & 0.98 \end{bmatrix}$$

Justification: The principal component transformation decorrelates the data, resulting in a diagonal covariance matrix. The diagonal entries are the eigenvalues corresponding to the selected principal components (in order: λ_3 for the first PC and λ_1 for the second PC). The off-diagonal elements are zero, indicating that the principal components are uncorrelated.

✓ On-Time (0.8 points) + Notebook PDF (0.2 points)

Submit your assignment as a PDF file before the deadline.

Submit Your Solution

Confirm that you've successfully completed the assignment.

Along with the Notebook, include a PDF of the notebook with your solutions.

`add` and `commit` the final version of your work, and `push` your code to your GitHub repository.

Submit the URL of your GitHub Repository as your assignment submission on Canvas.
